

Vatsal Vaghasiya

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PROFESSIONAL SUMMARY

MTech Data Science student and aspiring AI/ML engineer with industry internship experience and a first-author research paper under review on efficient LLM preference classification. Experienced in building and deploying ML models across LLMs, RAG, computer vision, and MLOps. 1st place winner at the Karnataka Education Datathon 2026.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Machine Learning and Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, NLP, computer vision, Transformers, CNNs, DeBERTa, CLIP, model calibration, OpenCV, MediaPipe

Generative AI and LLMs: retrieval-augmented generation (RAG), embeddings, vector search (ChromaDB), prompt engineering, LLM evaluation, Whisper ASR, Hugging Face

MLOps and Deployment: FastAPI, REST APIs, Docker, Docker Compose, CI/CD with GitHub Actions, model monitoring, Hugging Face Spaces, Cloudflare Workers, Git

Data and Analytics: pandas, NumPy, SciPy, ETL, feature engineering, OLS regression, clustering (K-Means, DBSCAN), Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

Data Science Intern | Amar Infotech (IT services), Ahmedabad, India | Jan 2025 - Jun 2025

- Built Python data processing pipelines handling 50,000+ records, improving data cleanliness and consistency by 35%.
- Trained predictive ML models with TensorFlow and scikit-learn to 87% accuracy, reducing false positives by 22%.
- Automated data extraction and validation workflows, cutting recurring manual processing time by 40%.
- Collaborated with engineering team to structure reporting tables and document data validation rules for production handoff.

PUBLICATIONS

Efficient LLM Preference Classification via Position Bias Mitigation | First author | Under review, 2026

- Built a 71M-parameter Siamese DeBERTa-v3 model that predicts human chatbot preference at 98% of state-of-the-art quality with 127x fewer parameters than 9B+ winning systems.
- Swap augmentation and post-hoc temperature scaling cut log loss 6.2% to 0.9871 and added 3.5 accuracy points; trained in 8.4 hours on free-tier T4 GPUs at \$0 cost.

AWARDS

1st Place, Karnataka Education Datathon 2026 | Akshara Foundation - Analysed arithmetic learning outcomes for 1.38M students across 31 districts, evaluated policy hypotheses via variance decomposition and regression, and isolated a procedural carry-and-borrow step gap as the primary driver of student difficulty.

PROJECTS

IPL Match Predictor, end-to-end MLOps pipeline | XGBoost, FastAPI, Docker Compose - Deployed three containerised services over 9 engineered features: a prediction REST API, a Population Stability Index data distribution monitor running every 5 minutes, and a live health dashboard.

Cachy, short-form media knowledge engine | FastAPI, Flutter, Whisper, LLM chain, SQLite - Turns video and articles into searchable knowledge cards via Whisper transcription, OCR and a three-provider LLM fallback chain; diagnosed a polling bug draining monthly database compute quota in 17 days, cutting queries 99.7%.

Constitution of India, RAG question answering | sentence-transformers, ChromaDB, Mistral-7B - Hand-wrote the full RAG loop in roughly 60 lines without LangChain, returning cited answers; a 20-query adversarial evaluation lifted accuracy from 65% to 78% by fixing chunk boundaries.

Bangalore Air Quality Hotspot Analysis | pandas, scikit-learn, DBSCAN - Consolidated a year of data from 13 CPCB stations and engineered 6 domain features; DBSCAN isolated critical pollution hotspots (Silk Board, peak AQI 500) that city-wide averages obscured.

Plant Disease Detector and CLIP Multimodal Search | PyTorch, CLIP, Hugging Face Spaces - Trained a CNN over 15 disease classes with visual feature attribution overlays for explainability; built CLIP text-to-image and image-to-image retrieval with a per-query category breakdown chart.

EDUCATION

M.Tech, Data Science | M.S. Ramaiah University of Applied Sciences, Bengaluru | Nov 2025 - Nov 2027 (expected)

B.Tech, Computer Engineering | SAL College of Engineering, Ahmedabad | Aug 2021 - May 2025 | CGPA 8.3 / 10.0

CERTIFICATIONS

IBM Data Science Professional Certificate (Coursera), 5 courses completed